

Building Intelligent Retrieval Augmented AI Systems

LLMs and vector databases work together to create intelligent AI systems that understand data.



Large language models (LLMs) are a foundational building block for modern AI applications. They use extensive text data for training, which enables them to complete various tasks like answering questions, creating summaries, translating languages and producing computer code. The models use deep language comprehension to produce responses that are relevant to the given situation.

Despite their advanced abilities, LLMs still remain imperfect. They function according to the training data, which restricts their performance. Their knowledge base remains fixed because they need external sources of information to learn about recent updates. The system generates hallucinations producing confident incorrect outputs, and cannot access confidential information or the

specialised knowledge businesses require for their daily operations.

Developers turn to orchestration frameworks such as LangChain to solve these problems. The system offers developers a standardised method to link LLM systems with external software tools, application programming interfaces and data processing systems. It enables users to create multi-step processes that include data collection and processing and response creation instead of using basic model prompts.

Another essential concept is the vector database which enables semantic search. Traditional databases rely on exact matches or keyword-based queries. Vector databases maintain their data as embeddings that represent text in numerical form to preserve its semantic meaning. Systems use similarity-based information retrieval to find information instead of matching exact phrases.

Specialised models create embeddings that transform text into vectors. Researchers can perform effective searches on the stored embeddings throughout the vector database by using similarity metrics, which include cosine similarity. Intelligent retrieval systems depend on this technology.

The combination of LLMs with LangChain and vector databases creates a powerful solution, which enables the following functions:

- LLMs perform reasoning functions that help them to create language content.
- Vector databases handle memory operations together with data retrieval tasks.
- LangChain manages the complete system by coordinating all components.

This system enables applications to produce specific answers that utilise